Adults income

Data mining project 1st semester 1447h

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PROJECT TEAM

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PROBLEM

- Income inequality is a growing problem
- Not understanding why some people earn more than $50K and others earn less
- Many personal and demographic factors affect income
- Hard to see these patterns without analyzing real data
- need a clearer view

DATA

- **The dataset**: Adult Income dataset, from Kaggle.
- **contains**: demographic, educational, and employment-related information about individuals.
- **commonly used**: to study income prediction.

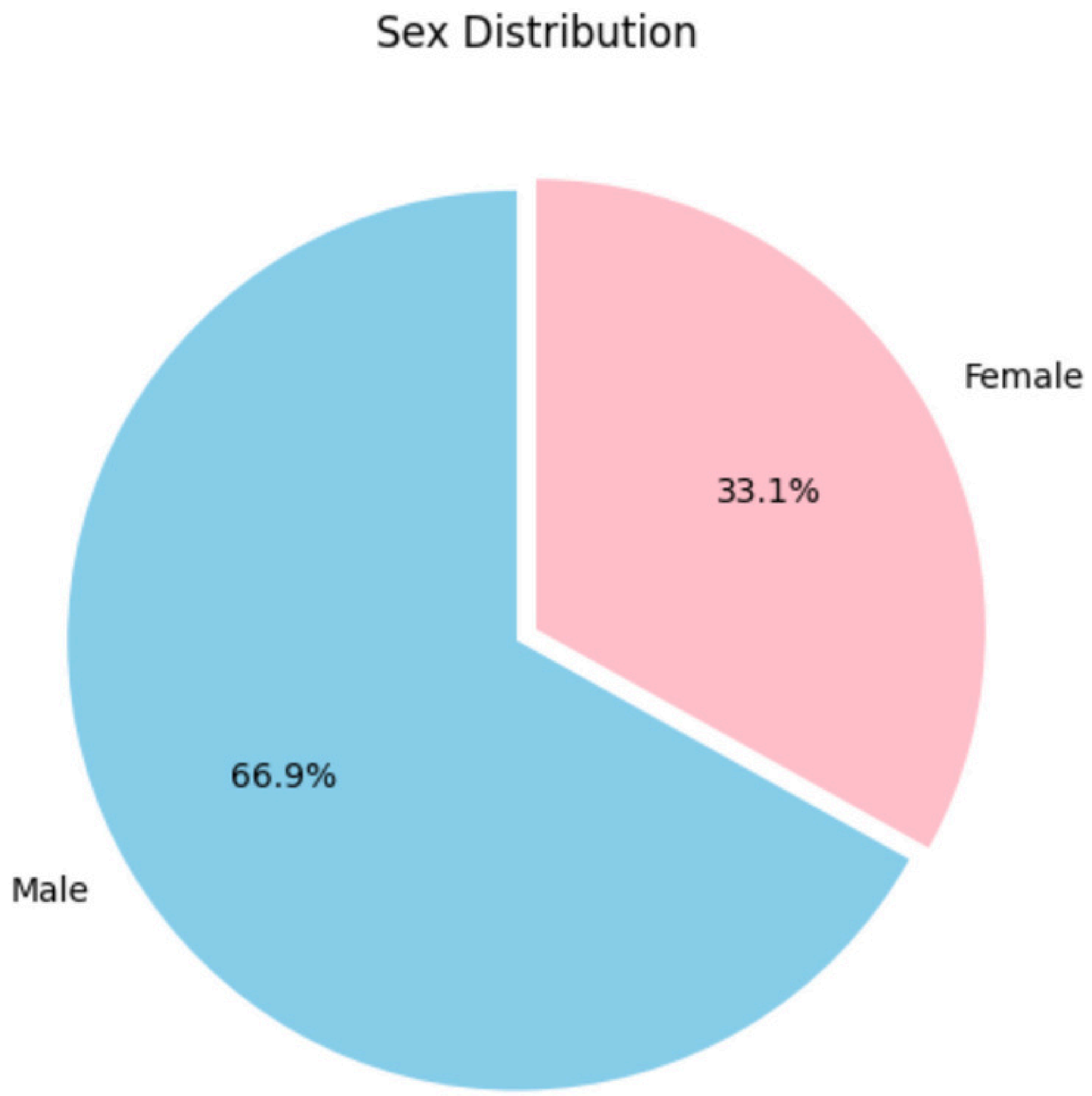
PROJECT GOAL

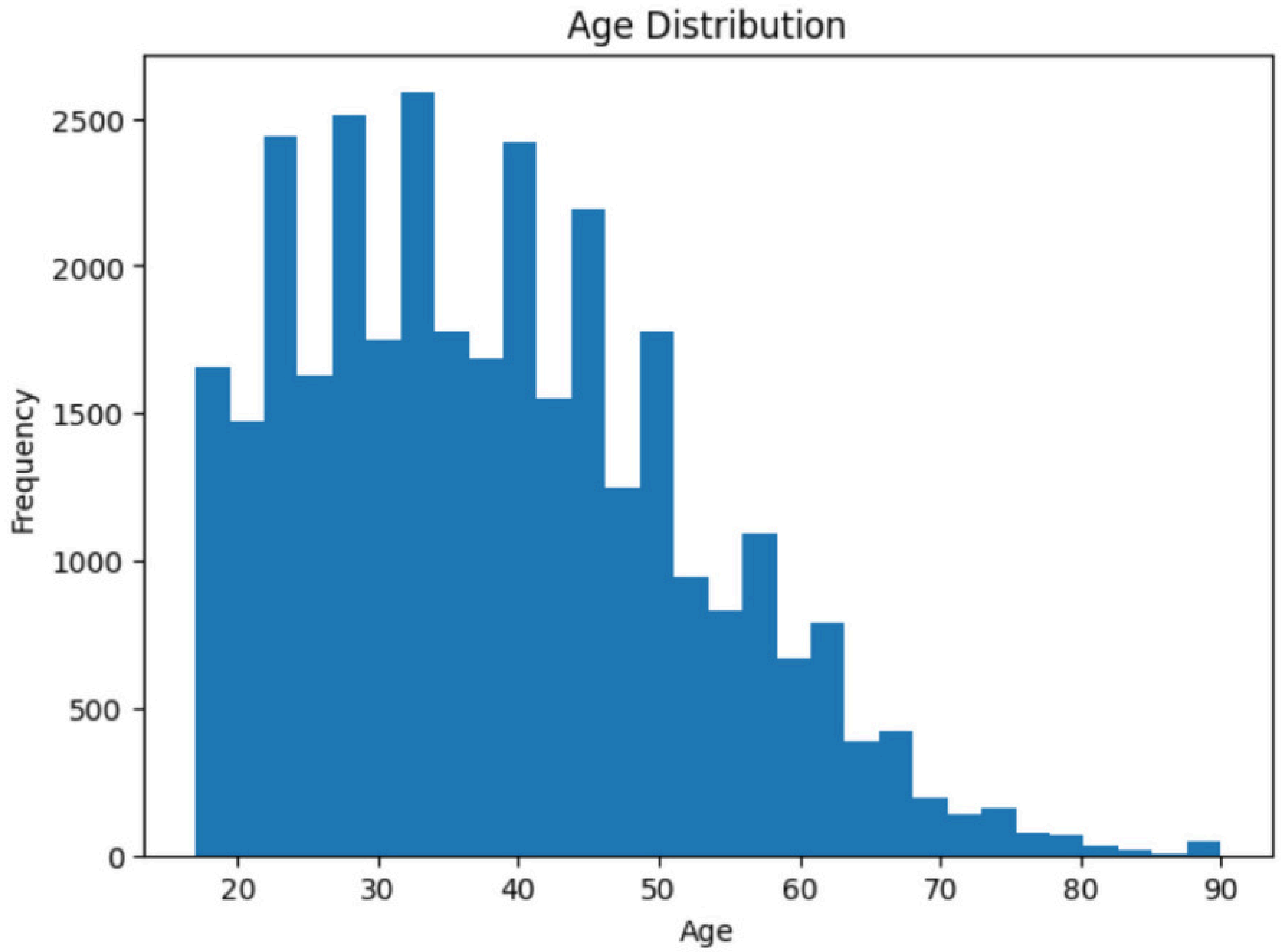
- Identify the key demographic and personal factors that influence whether an adult earns more or less than $50K per year.
- Build classification and clustering models to predict income levels and group individuals with similar socioeconomic characteristics.

DATA DICTIONARY

	Attribute Name	Description	Data Type	Possible Values
0	age	Age of the individual	Numeric	17–90
1	workclass	Type of employer	Categorical	Private, Self-emp-not-inc, Local-gov, ?, etc.
2	fnlwgt	Census weight assigned to the individual	Numeric	Large integer values
3	education	Highest education level	Categorical	HS-grad, Bachelors, Masters, Some-college, etc.
4	education.num	Numeric representation of education level	Numeric	1–16
5	marital.status	Marital status	Categorical	Married, Divorced, Never-married, Widowed, etc.
6	occupation	Occupation type	Categorical	Sales, Exec-managerial, Tech-support, ?, etc.
7	relationship	Relationship within household	Categorical	Husband, Wife, Unmarried, Own-child, etc.
8	race	Racial background	Categorical	White, Black, Asian-Pac-Islander, Other
9	sex	Gender of the individual	Categorical	Male, Female
10	capital.gain	Investment gain	Numeric	0–99999
11	capital.loss	Investment loss	Numeric	0–4356
12	hours.per.week	Hours worked per week	Numeric	1–99
13	native.country	Country of origin	Categorical	United-States, Mexico, Germany, etc.
14	income	Income category (target variable)	Binary	<=50K / >50K

DATA REPRESENTATION





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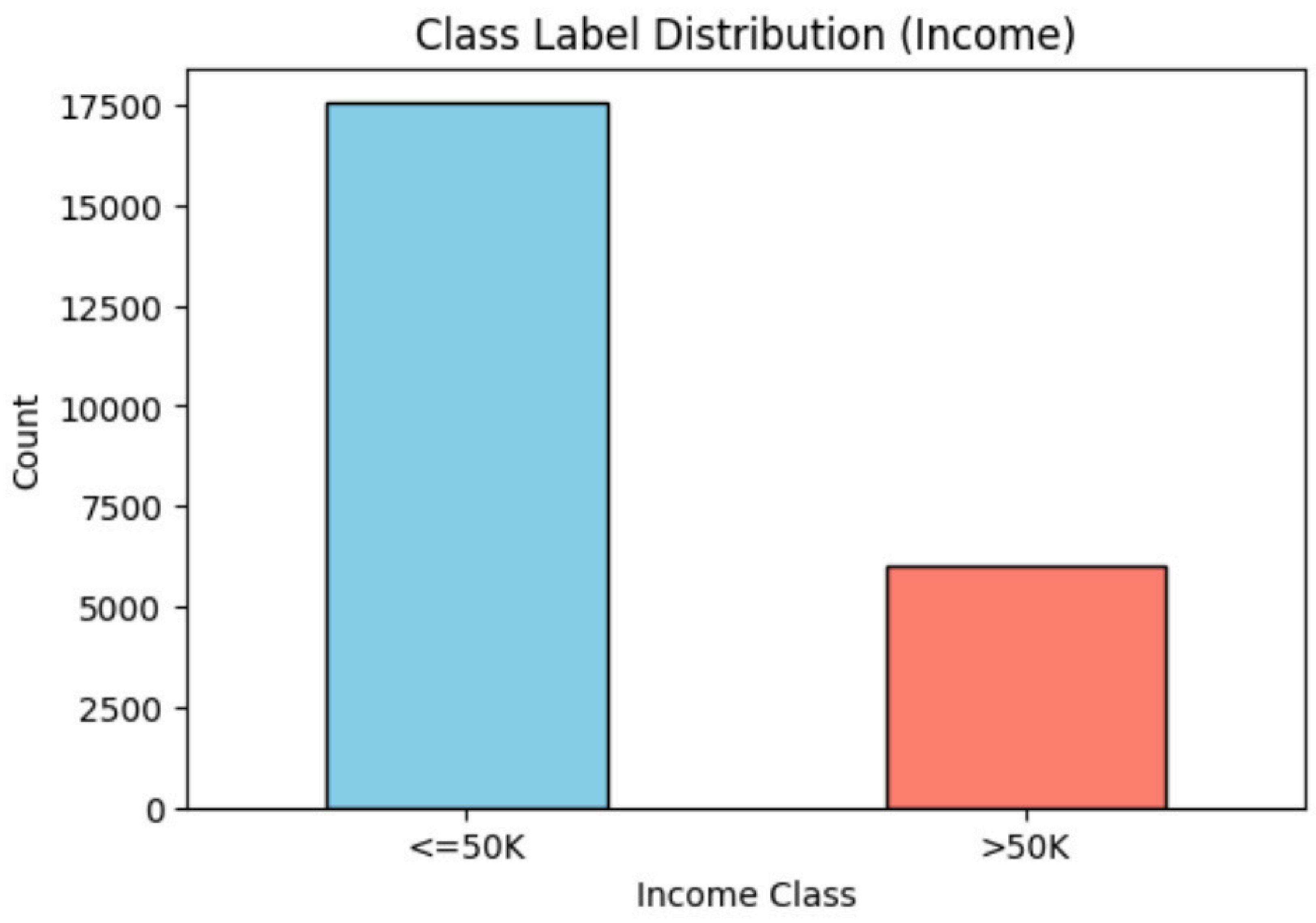

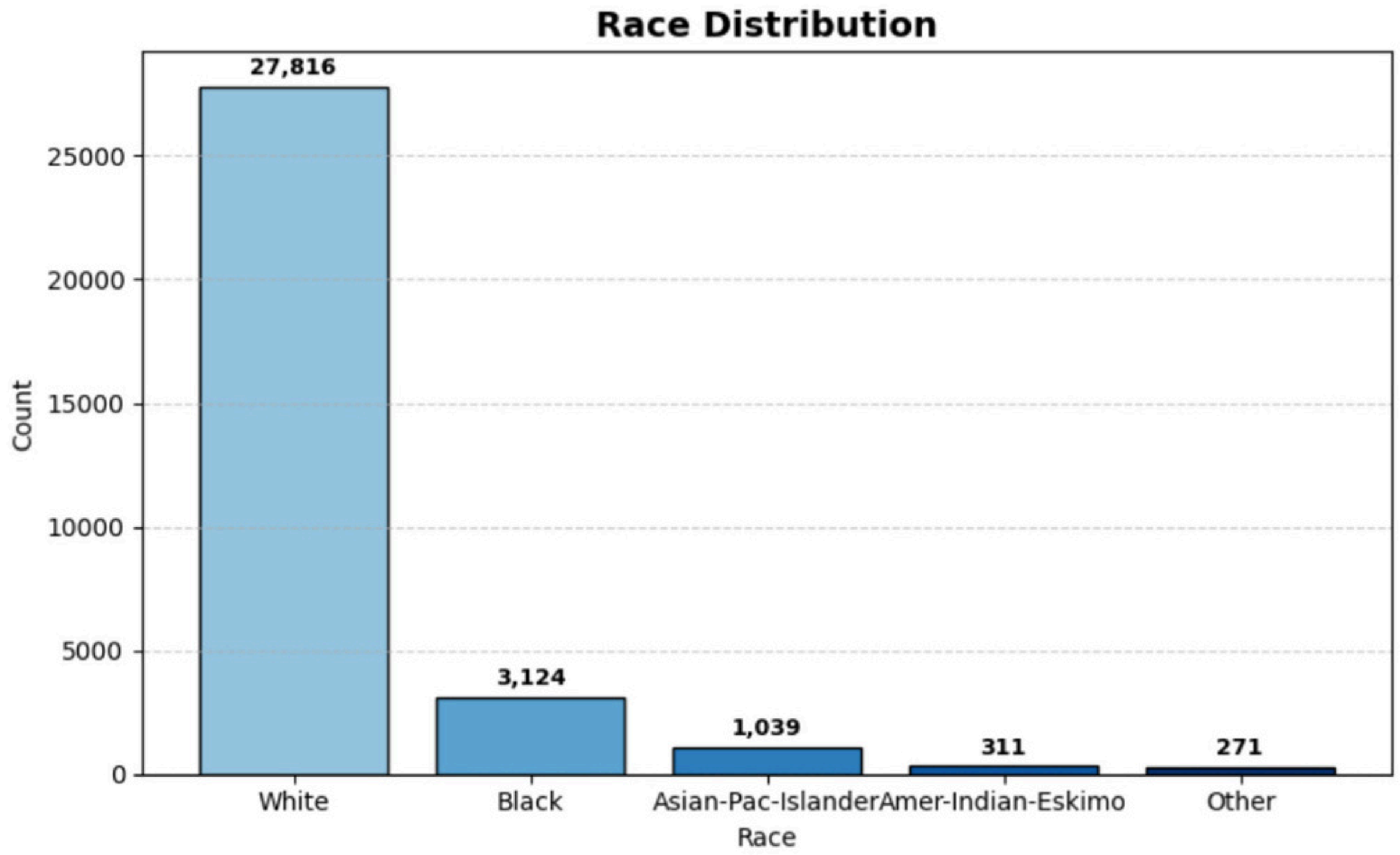

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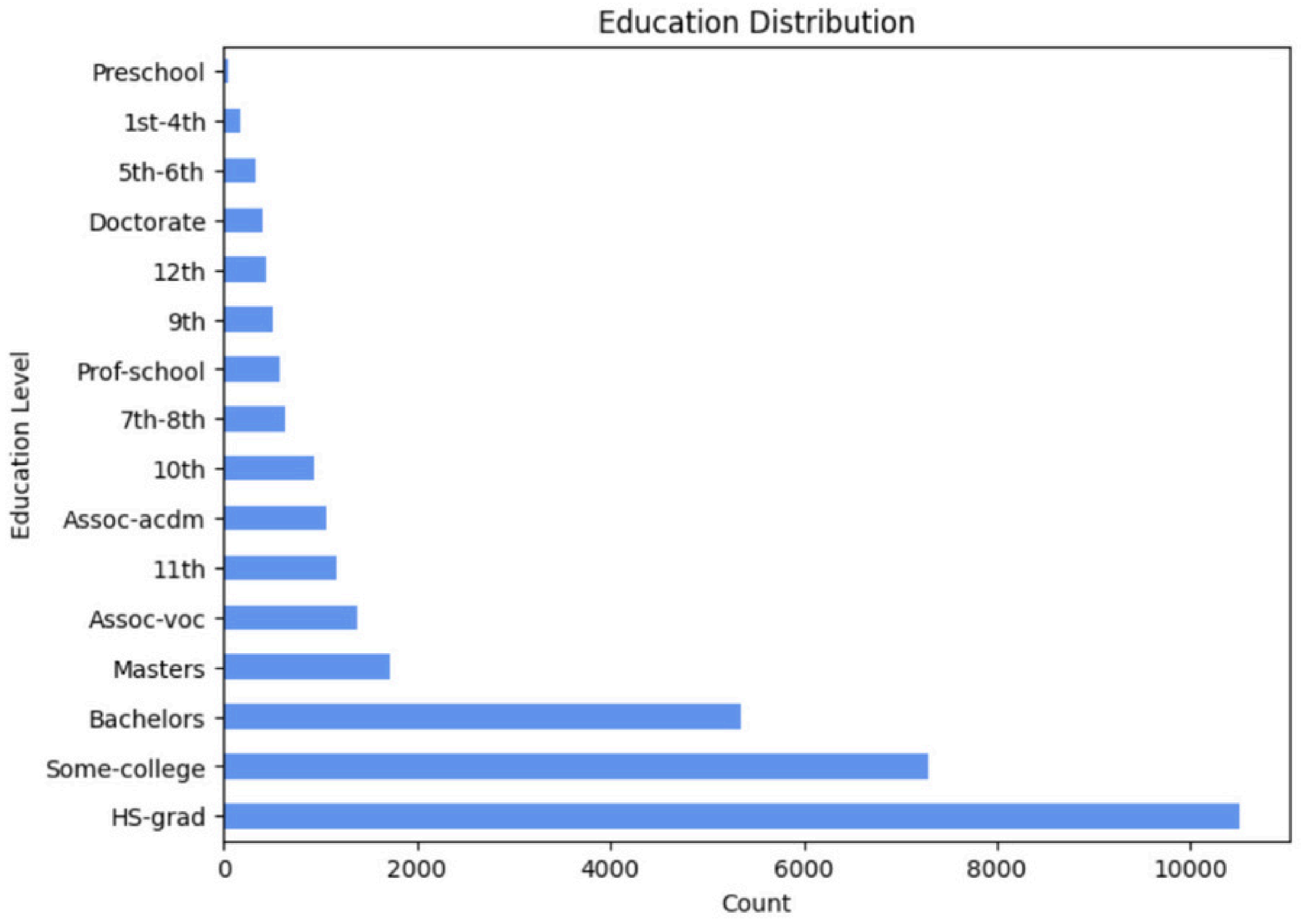

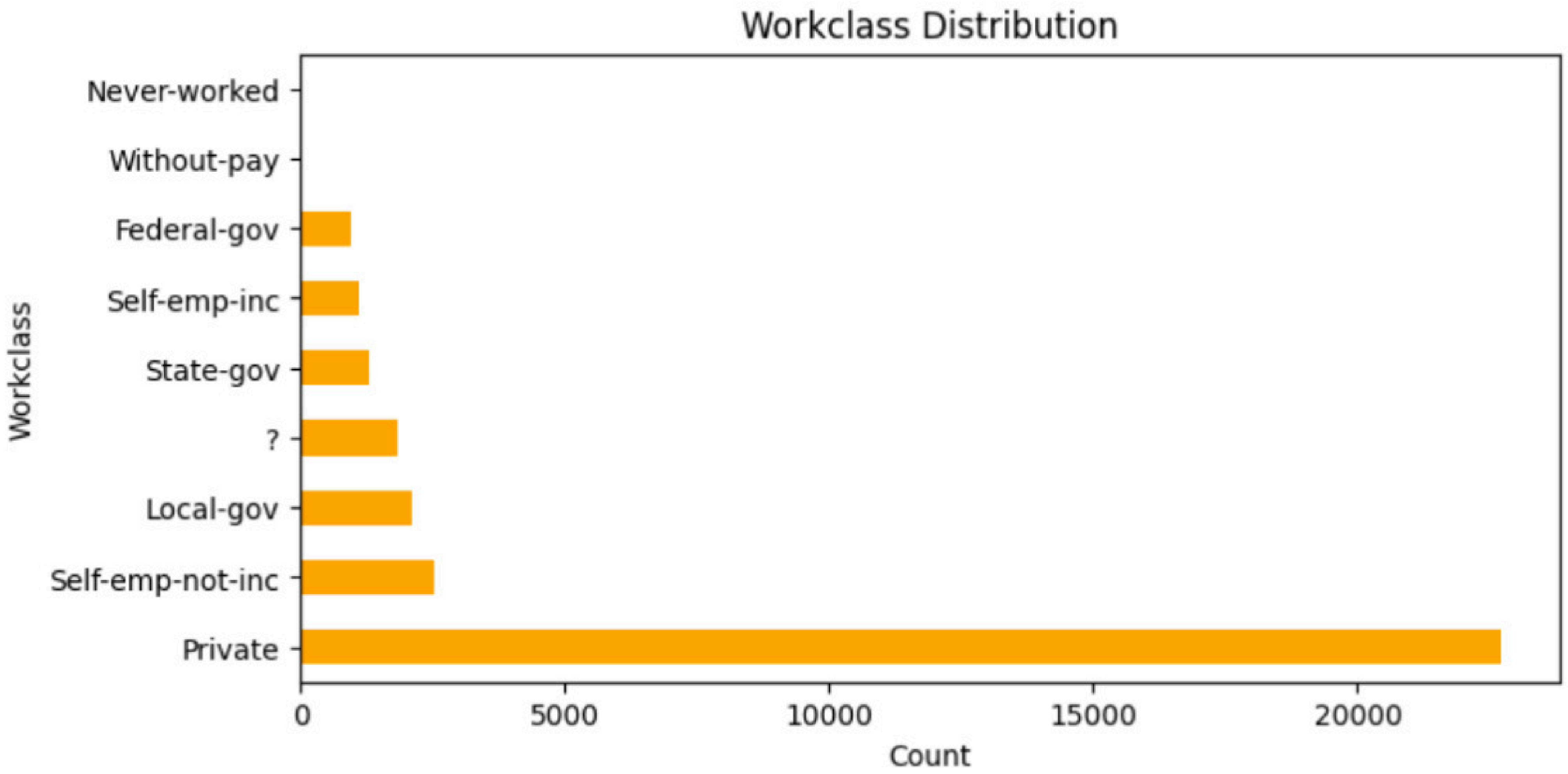

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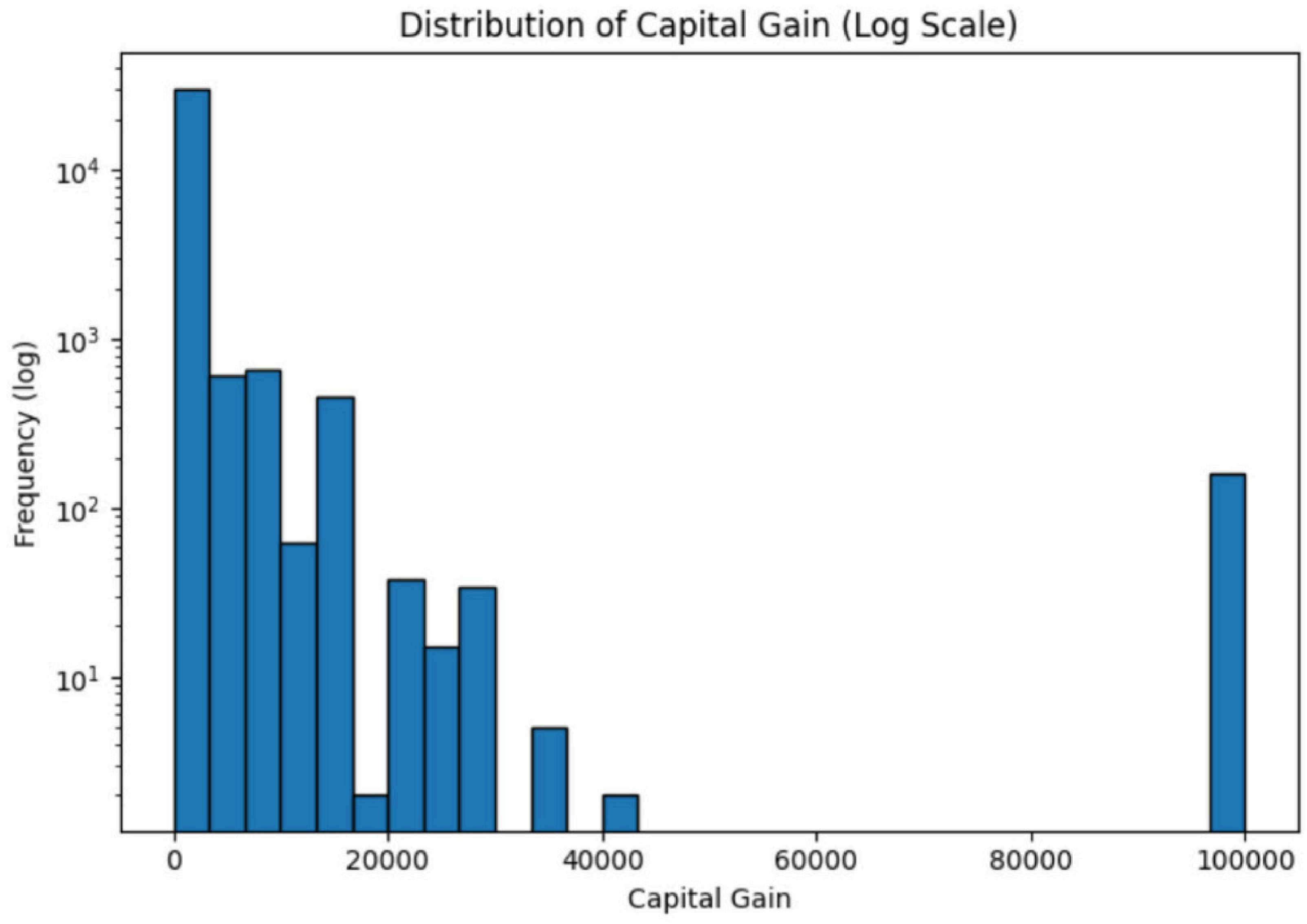

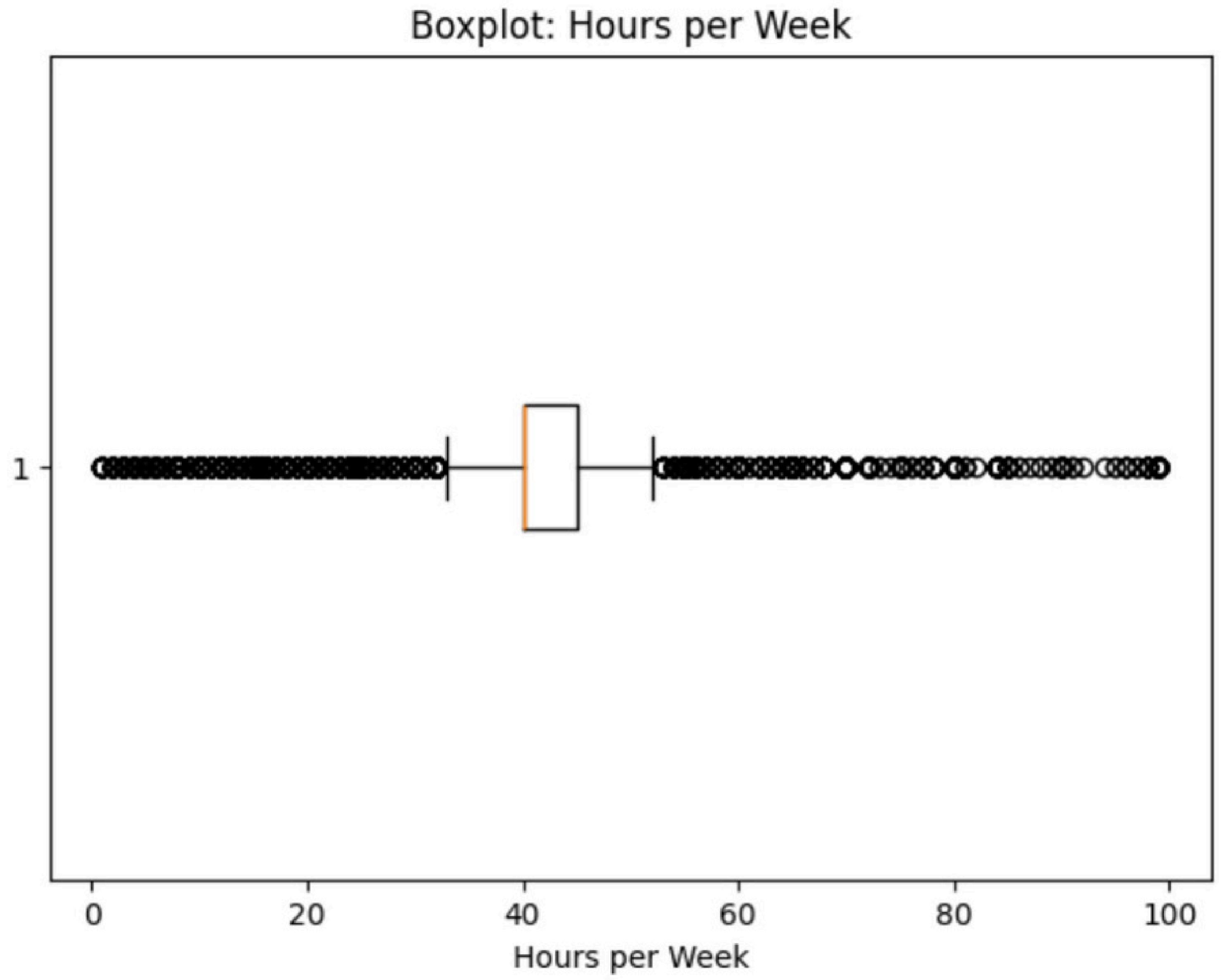

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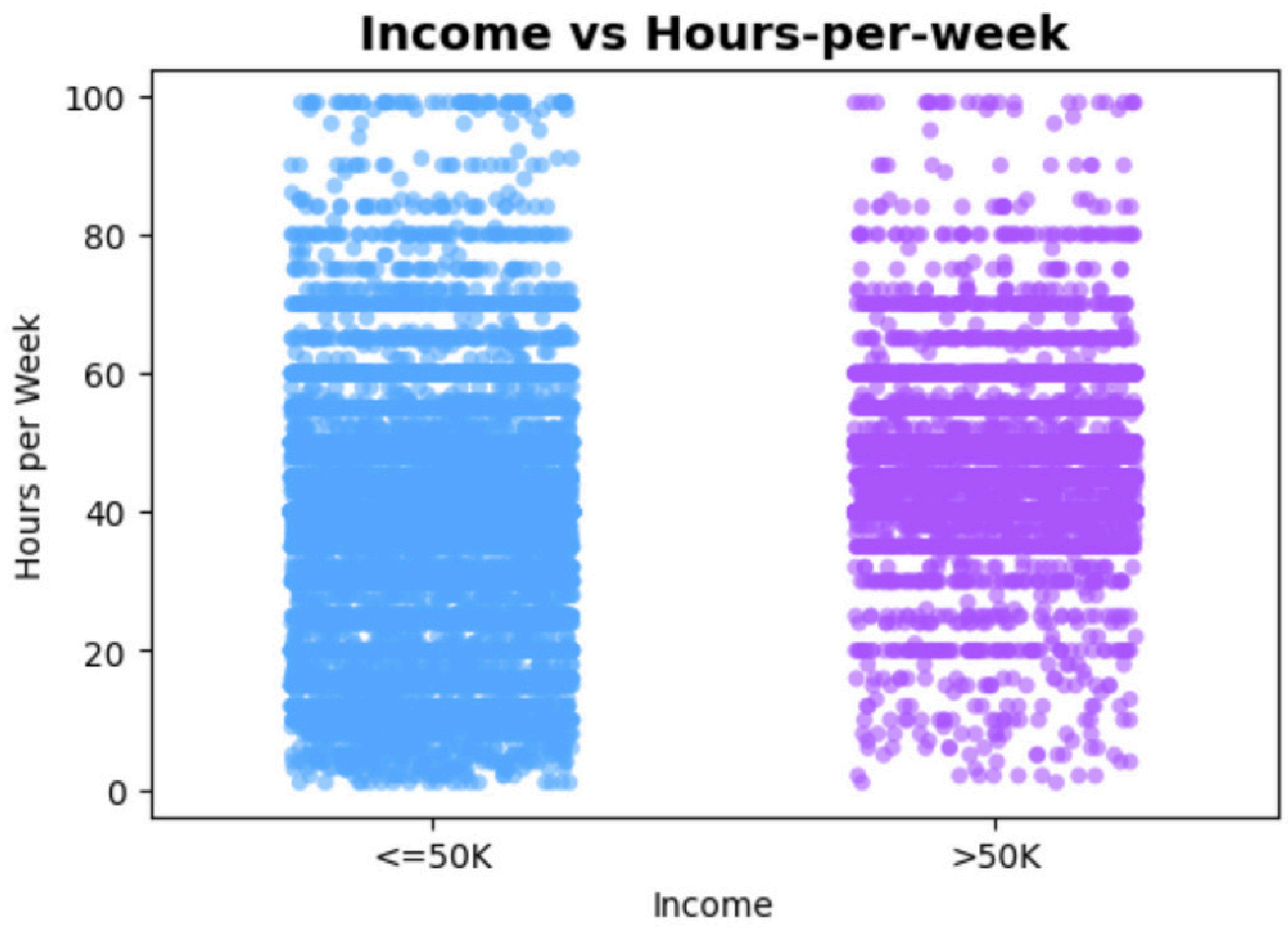

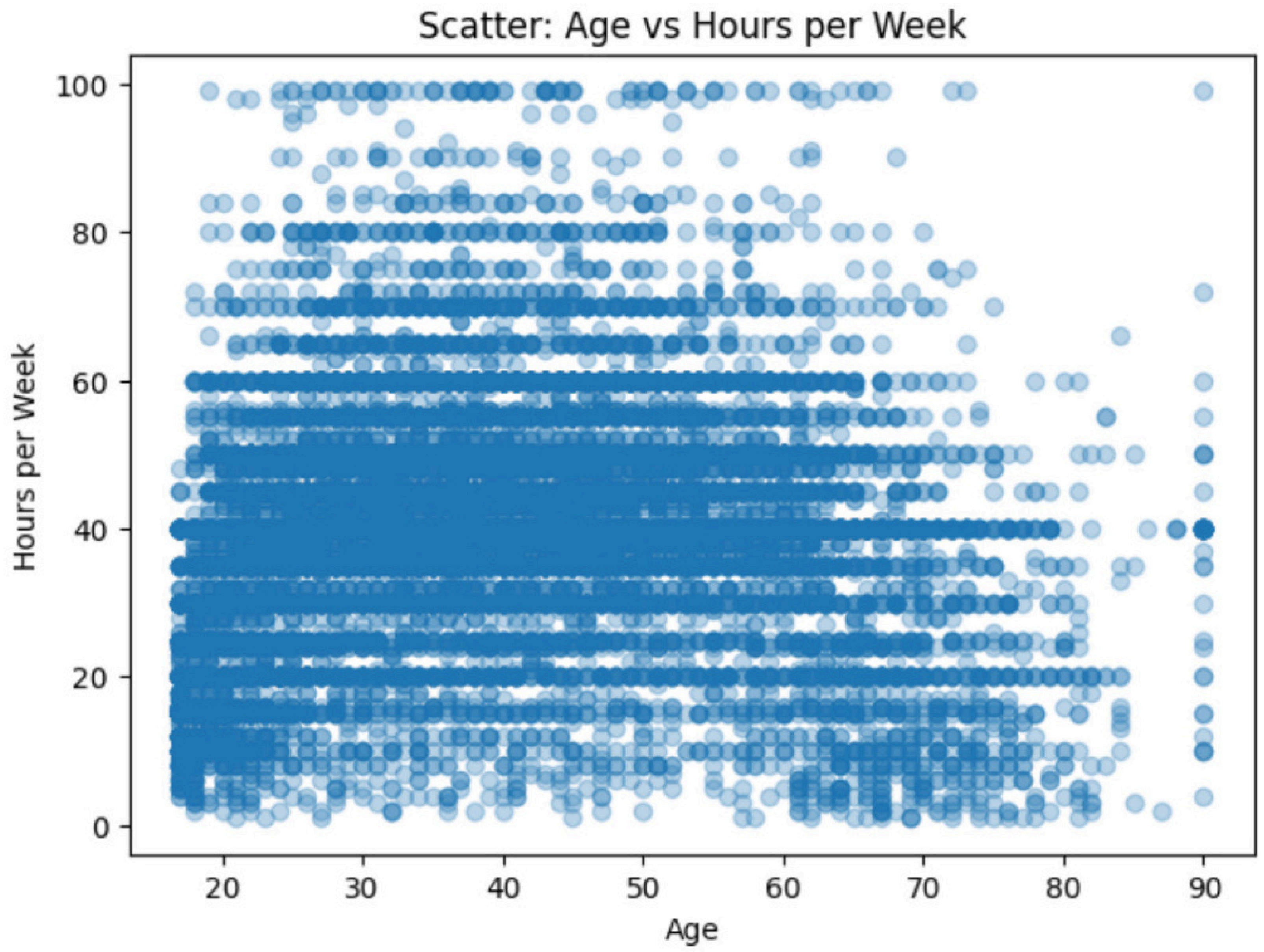

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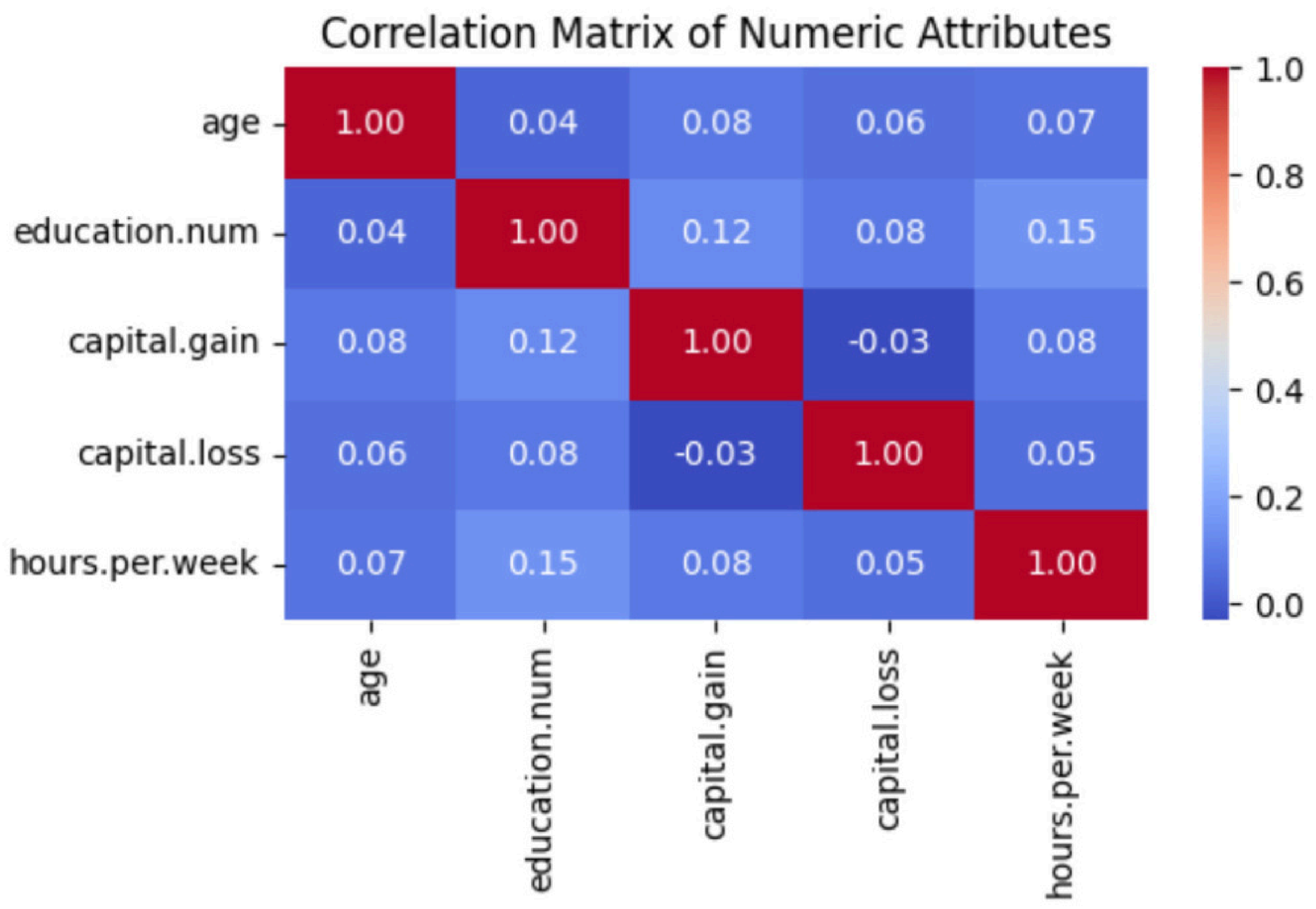


DATA PREPROCESSING

- **Data cleaning**: Missing values, Noisy data.
- **Data Transformation**: Encoding, Normalization, Discretization.
- **Data reduction**: Attribute subset selection.

DATA CLEANING

- Detect any outliers and remove them (using IQR)

```
Q1 = df["hours.per.week"].quantile(0.25)
Q3 = df["hours.per.week"].quantile(0.75)
IQR = Q3 - Q1
df = df[(df["hours.per.week"] >= Q1 - 1.5*IQR) & (df["hours.per.week"] <= Q3 + 1.5*IQR
```

DATA CLEANING

- Find missing values and handle them (replaced with mode)

```
df = df.replace("?", pd.NA)
```

```
cols_with_missing = ["workclass", "occupation", "native.country"]

for col in cols_with_missing:
    mode_value = df[col].mode()[0]
    df[col] = df[col].fillna(mode_value)
```

DATA TRANSFORMATION

Encoding

One-Hot Encoding - Multi-category coulmns

```
cols_to_encode = ["workclass", "occupation", "relationship"]
cols_exist = [col for col in cols_to_encode if col in df.columns]

if cols_exist:
    df = pd.get_dummies(df, columns=cols_exist, drop_first=True)
    print(f" Applied One-Hot Encoding to: {cols_exist}")
else:
    print(" Columns not found. Available columns:")
    print(df.columns.tolist())
```

```
Applied One-Hot Encoding to: ['workclass', 'occupation', 'relationship']
```

```
print("Columns after encoding:\n", df.columns.tolist())
```

Label Encoding - Gender

```
le = LabelEncoder()
df["sex"] = le.fit_transform(df["sex"])
```

```
print(df["sex"].unique())
```

```
[0 1]
```

DATA TRANSFORMATION

- Normaliztion

Normalization - Min-Max Scaling

```
numeric_cols = ["age", "hours.per.week", "capital.gain", "capital.loss"]

scaler = MinMaxScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
```

```
print(df[numeric_cols].min().round(2))
print(df[numeric_cols].max().round(2))
```

```
age               0.0
hours.per.week    0.0
capital.gain      0.0
capital.loss      0.0
dtype: float64
age               1.0
hours.per.week    1.0
capital.gain      1.0
capital.loss      1.0
dtype: float64
```

DATA TRANSFORMATION

- Discretization

```
from sklearn.preprocessing import KBinsDiscretizer

discretizer = KBinsDiscretizer(n_bins=4, encode='ordinal', strategy='uniform')

df["age_binned"] = discretizer.fit_transform(df[["age"]])
df["hours_binned"] = discretizer.fit_transform(df[["hours.per.week"]])

# Show a sample
df[["age", "age_binned", "hours.per.week", "hours_binned"]].head(10)
```

	age	age_binned	hours.per.week	hours_binned
0	1.000000	3.0	0.368421	1.0
2	0.671233	2.0	0.368421	1.0
3	0.506849	2.0	0.368421	1.0
4	0.328767	1.0	0.368421	1.0
5	0.232877	0.0	0.631579	2.0
6	0.287671	1.0	0.368421	1.0
8	0.698630	2.0	0.368421	1.0
10	0.383562	1.0	0.105263	0.0
11	0.287671	1.0	0.631579	2.0
14	0.465753	1.0	0.368421	1.0

```
display(df[['age', 'age_binned', 'hours.per.week', 'hours_binned', 'income']].head(3))
```

	age	age_binned	hours.per.week	hours_binned	income
0	1.000000	3.0	0.368421	1.0	<=50K
2	0.671233	2.0	0.368421	1.0	<=50K
3	0.506849	2.0	0.368421	1.0	<=50K

DATA REDUCTION

Feature selection

```python
from sklearn.feature_selection import SelectKBest, chi2
import pandas as pd

X = df.drop("income", axis=1)
y = df["income"]

X = pd.get_dummies(X, drop_first=True)

selector = SelectKBest(score_func=chi2, k=10)
X_new = selector.fit_transform(X, y)

selected_columns = X.columns[selector.get_support()]
print("Top 10 selected features:")
print(selected_columns)
```

```
Top 10 selected features:
Index(['fnlwgt', 'education.num', 'occupation_Exec-managerial',
       'occupation_Prof-specialty', 'relationship_Not-in-family',
       'relationship_Own-child', 'education_Bachelors', 'education_Masters',
       'marital.status_Married-civ-spouse', 'marital.status_Never-married'],
      dtype='object')
```

```python
for col, score in zip(selected_columns, selector.scores_):
    print(f"{col}: {score:.2f}")
```

```
fnlwgt: 137.94
education.num: 137105.43
occupation_Exec-managerial: 1590.35
occupation_Prof-specialty: 315.02
relationship_Not-in-family: 494.03
relationship_Own-child: 193.74
education_Bachelors: 89.29
education_Masters: 28.62
marital.status_Married-civ-spouse: 1.37
marital.status_Never-married: 75.16
```

PREPROCESSED DATA

Snapshots: Raw vs Preprocessed

```
print("Raw dataset sample:")
display(raw_df.head(3))

print("\nPreprocessed dataset sample (selected columns):")
display(df[['age', 'age_binned', 'hours.per.week', 'hours_binned', 'income']].head(3))
```

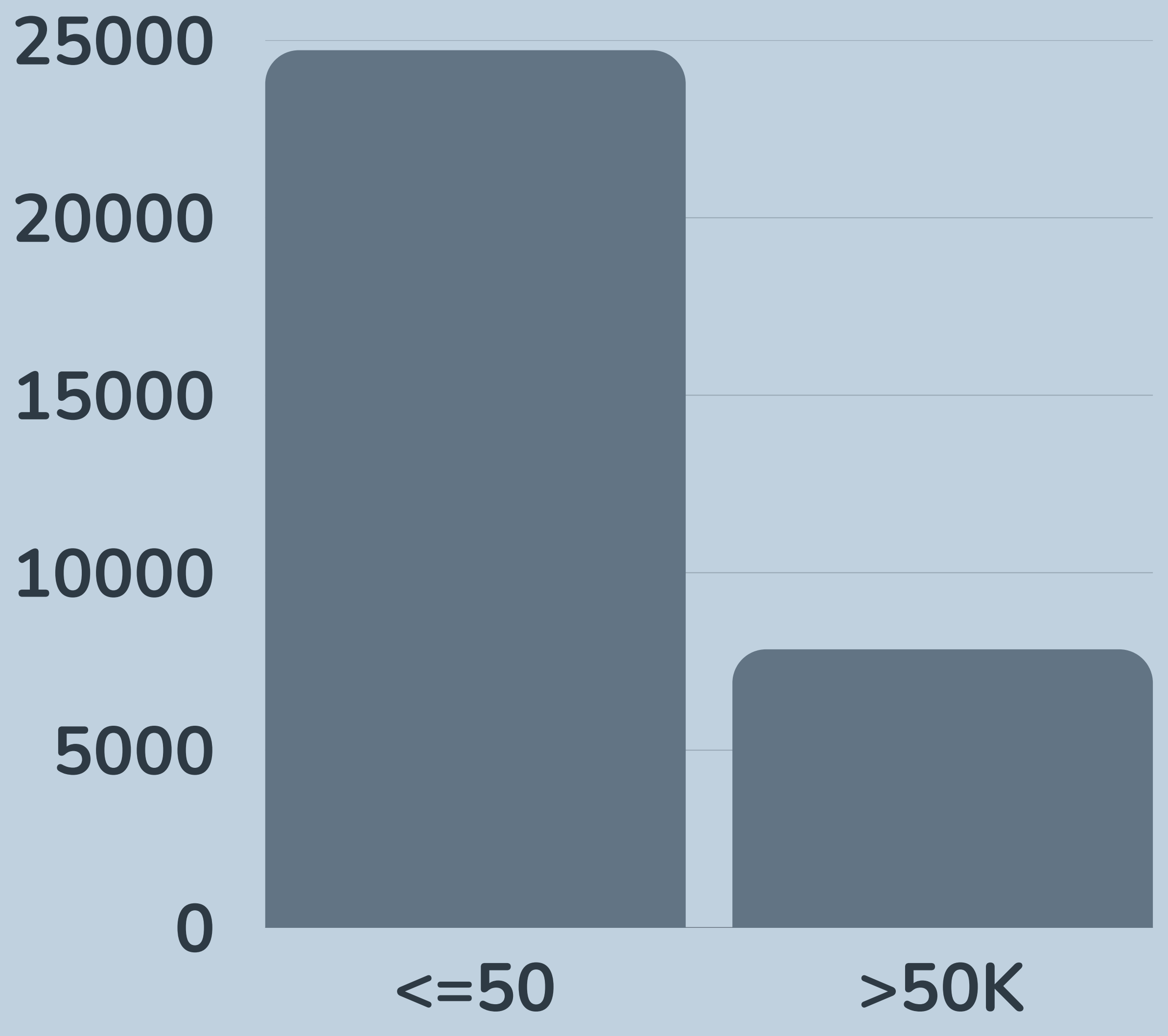
```
Raw dataset sample:
```

	age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationship	race
0	90	?	77053	HS-grad	9	Widowed	?	Not-in-family	White
1	82	Private	132870	HS-grad	9	Widowed	Exec-managerial	Not-in-family	White
2	66	?	186061	Some-college	10	Widowed	?	Unmarried	Black

```
Preprocessed dataset sample (selected columns):
```

	age	age_binned	hours.per.week	hours_binned	income
0	1.000000	3.0	0.368421	1.0	<=50K
2	0.671233	2.0	0.368421	1.0	<=50K
3	0.506849	2.0	0.368421	1.0	<=50K

UNBALANCED DATASET



- The income classes dataset are not balanced.
- Most individuals fall into the <=50K income category.
- The >50K class is significantly smaller.
- This imbalance affects model performance and evaluation.
- Accuracy is not enough, we rely on: precision, recall, F1-score, and confusion matrices.

CLASSIFICATION

Objective

- Predict whether income >50K or <=50K

Why classification

- Supervised model learns from patterns in training data

Target variable

- income (binary)

Algorithms used

- Decision Tree (Gini)
- Decision Tree (Entropy)

CLASSIFICATION PROCESS

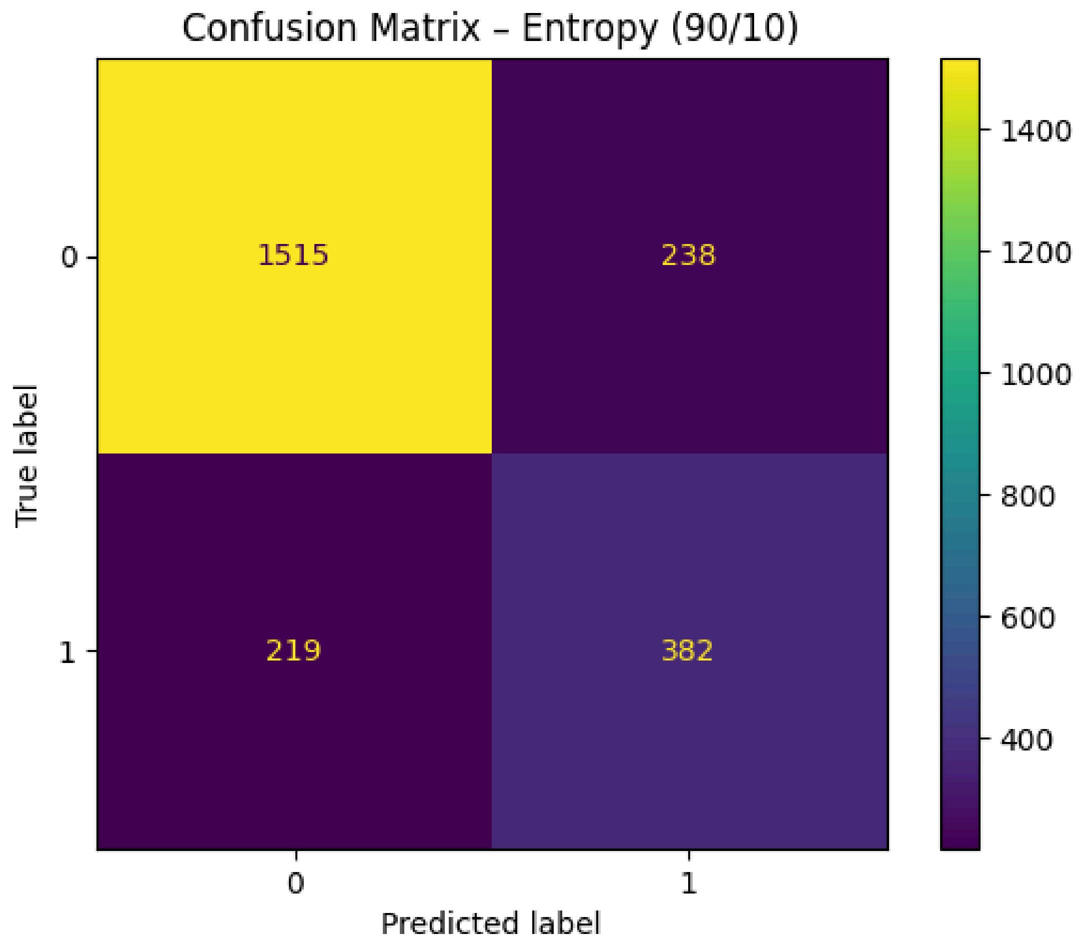
Steps

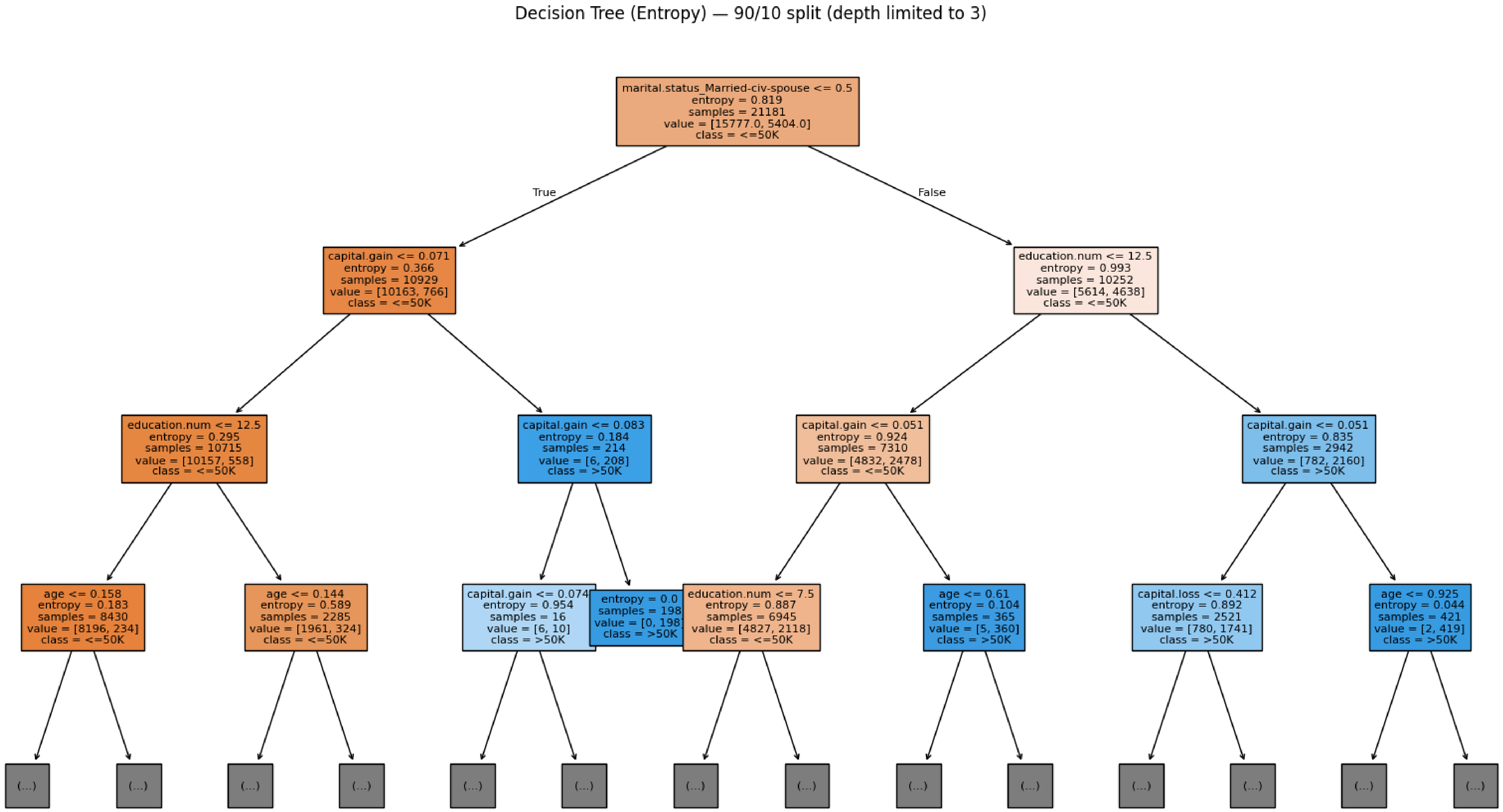
1. Split data into: 90/10, 80/20, 70/30
2. Train Decision Trees
3. Evaluate performance
4. Compare models

Metrics

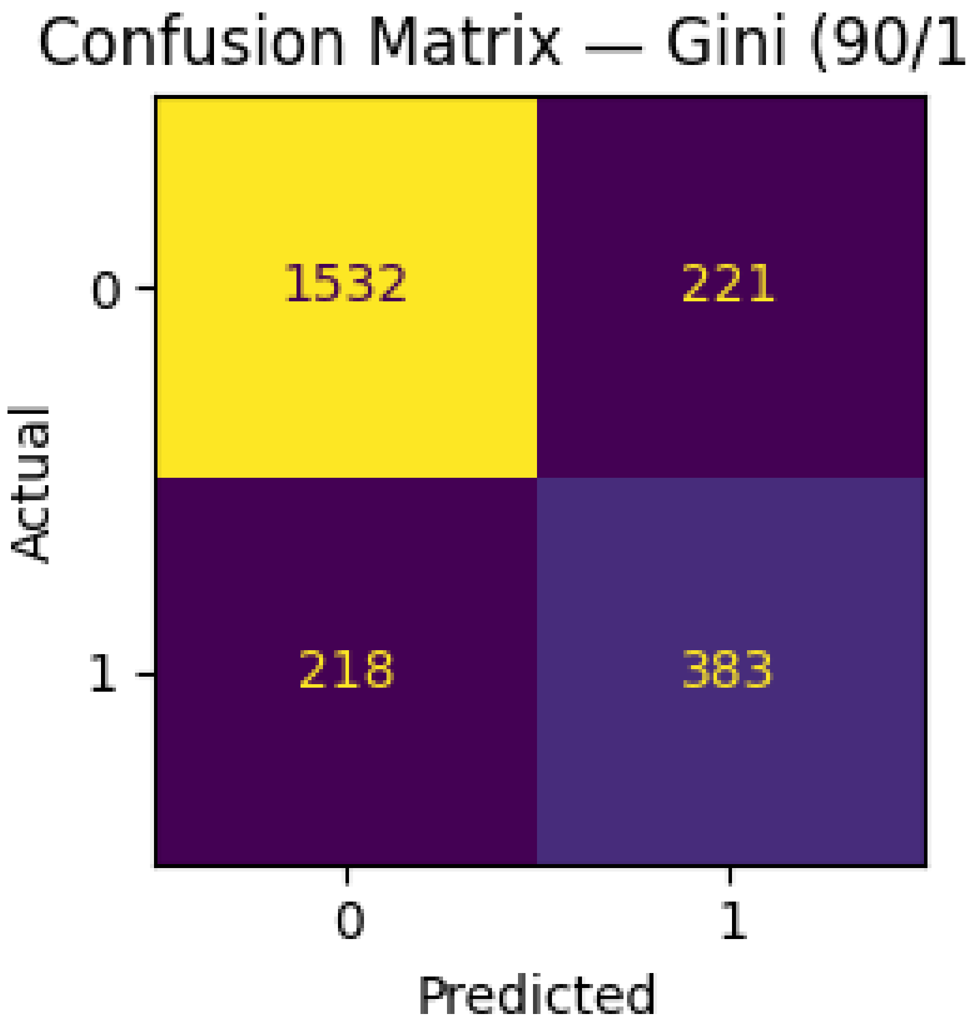
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

INFORMATION GAIN (ENTROPY)

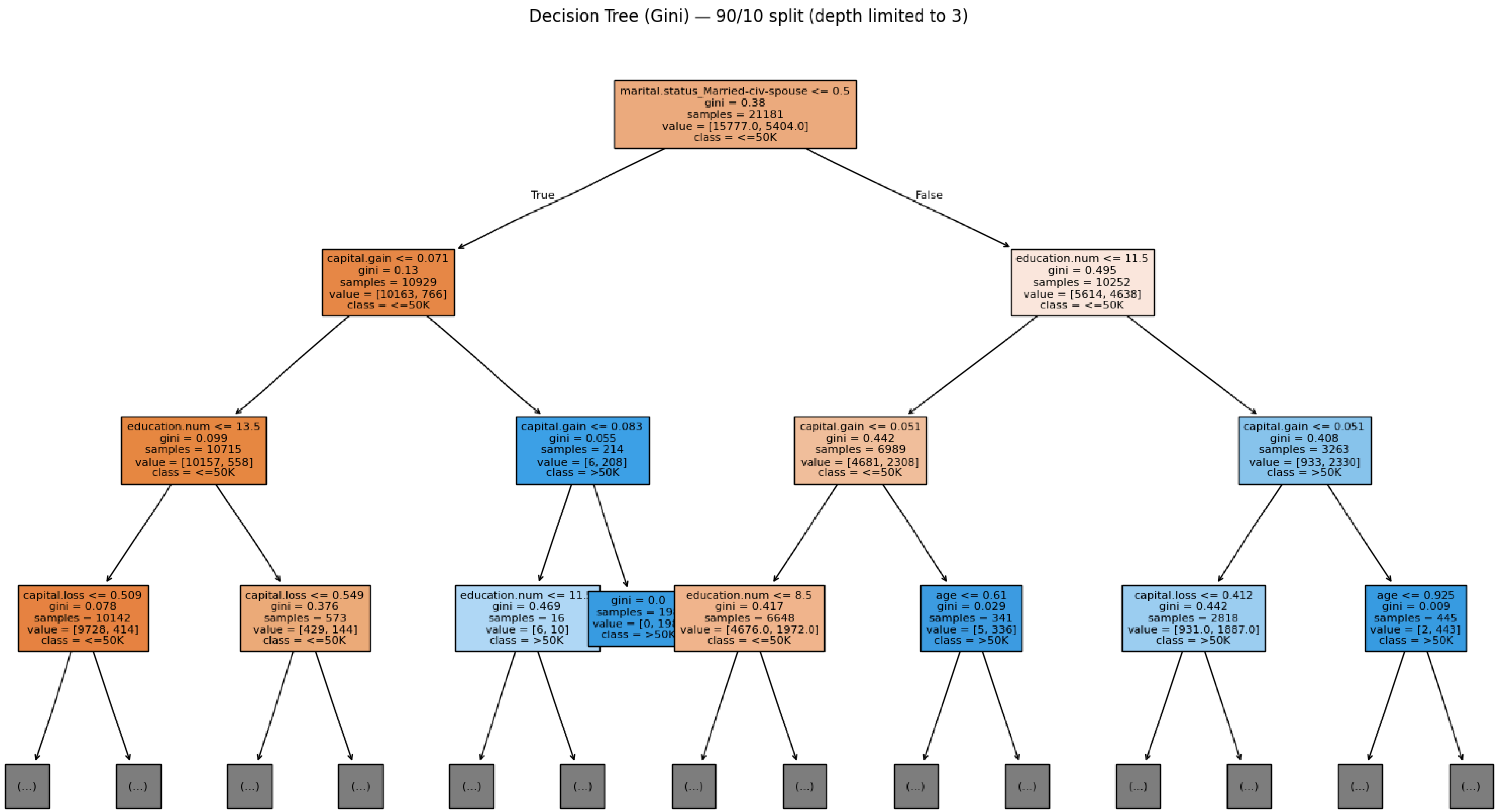




GINI INDEX



Confusion Matrix — Gini (90/1
0
1
Actual
1532
221
218
383
0
1
Predicted

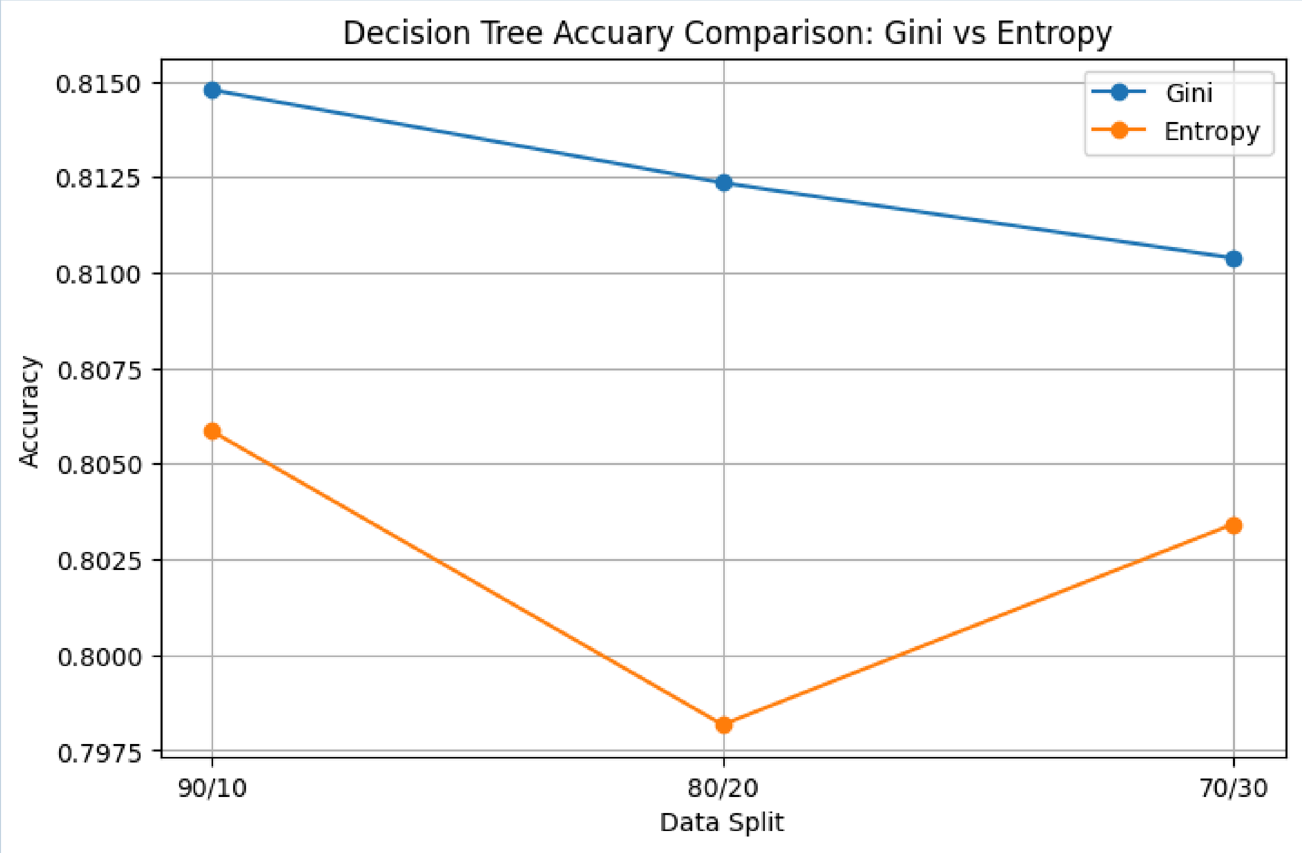


Decision Tree (Gini) — 90/10 split (depth limited to 3)
marital.status_Married-civ-spouse <= 0.5
gini = 0.38
samples = 21181
value = [15777.0, 5404.0]
class = <=50K
True
False
capital.gain <= 0.071
gini = 0.13
samples = 10929
value = [10163, 766]
class = <=50K
education.num <= 11.5
gini = 0.495
samples = 10252
value = [5614, 4638]
class = <=50K
education.num <= 13.5
gini = 0.099
samples = 10715
value = [10157, 558]
class = <=50K
capital.gain <= 0.083
gini = 0.055
samples = 214
value = [6, 208]
class = >50K
capital.gain <= 0.051
gini = 0.442
samples = 6989
value = [4681, 2308]
class = <=50K
capital.gain <= 0.051
gini = 0.408
samples = 3263
value = [933, 2330]
class = >50K
capital.loss <= 0.509
gini = 0.078
samples = 10142
value = [9728, 414]
class = <=50K
capital.loss <= 0.549
gini = 0.376
samples = 573
value = [429, 144]
class = <=50K
gini = 0.469
samples = 16
value = [6, 10]
class = >50K
gini = 0.0
education.num <= 8.5
gini = 0.417
samples = 6648
value = [4676.0, 1972.0]
class = <=50K
age <= 0.61
gini = 0.029
samples = 341
value = [5, 336]
class = >50K
capital.loss <= 0.412
gini = 0.442
samples = 2818
value = [931.0, 1887.0]
class = >50K
age <= 0.925
gini = 0.009
samples = 445
value = [2, 443]
class = >50K
(...)

CLASSIFICATION RESULTS

Aspect	Gini Index	Information Gain
Accuracy	Slightly lower (~0.80–0.82)	Slightly higher (~0.81–0.83)
Training Speed	Faster	Slightly slower
Model Depth	Shallower	Slightly deeper
Sensitivity to Purity	Moderate	Higher
Interpretability	Very good	Good
Best Split Criterion (for our dataset)	–	✅ Information Gain

CLASSIFICATION FINDINGS



- 90/10 achieved highest accuracy
- Entropy performed slightly better
- Results consistent across splits
- Tree interpretable and fast

Classification revealed:

- Income strongly influenced by:

1. Education
2. Marital status
3. Capital gain
4. Work hours

- Entropy > Gini

(slight numerical advantage)

- Tree interpretability is strong

CLUSTERING

Objective

- Identify natural groups of adults based on socio-economic features

Why clustering

- Unsupervised model finds hidden patterns without using income label.

Target variable

- Five numerical: age, education-num, hour per week, capital gain, captial loss.

Algorithms used

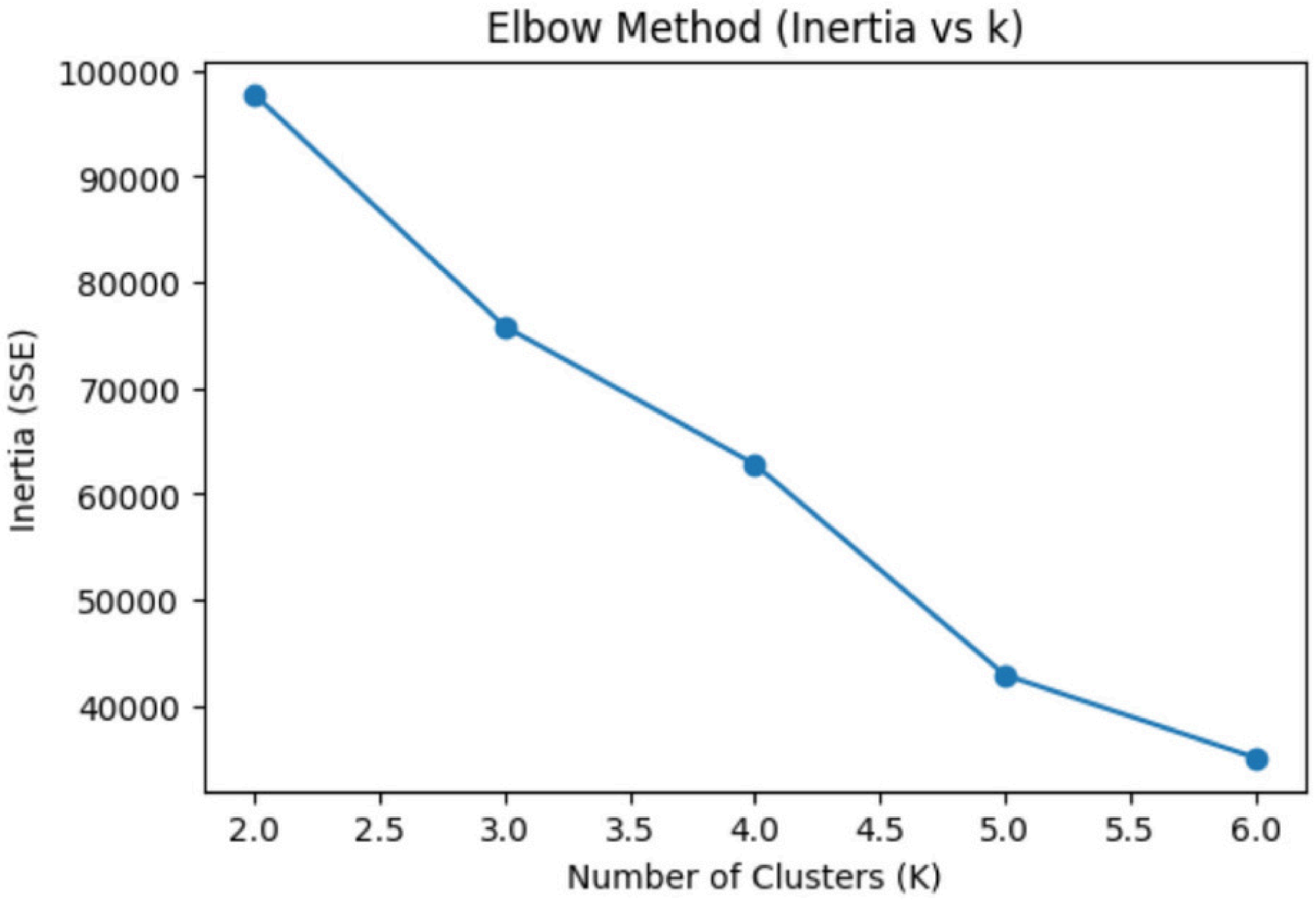
- K means (k=2 to 6)

Evaluation matrices

- Elbow Method (Inertia)
- Silhouette Score (cluster quality)

CLUSTERING EVALUATION

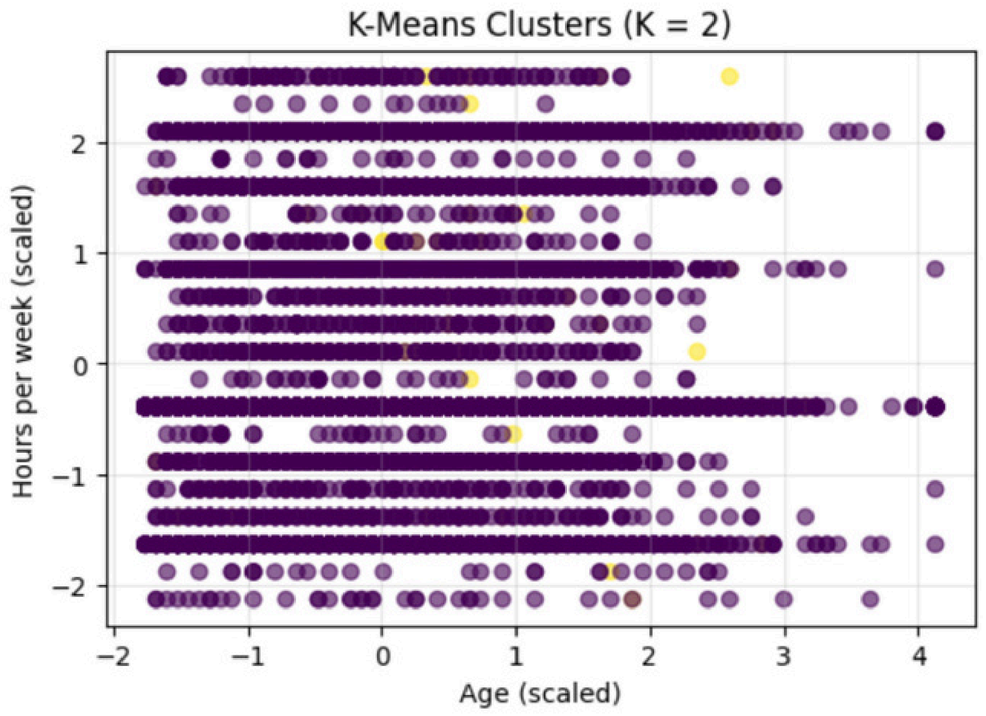
Elbow method



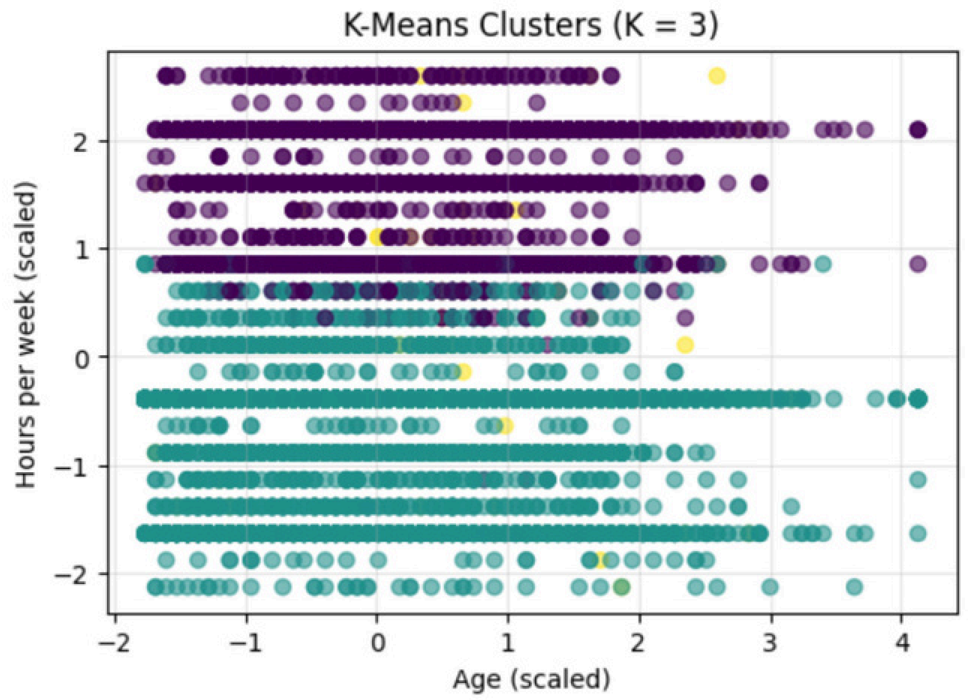
Silhouette score

CLUSTERING VISULIZE

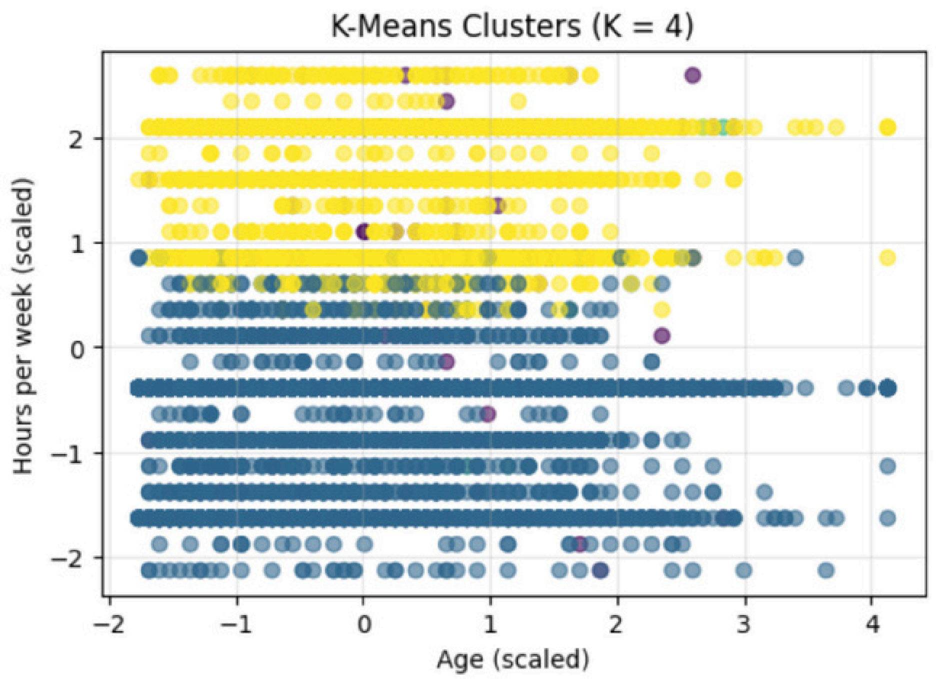
K = 2



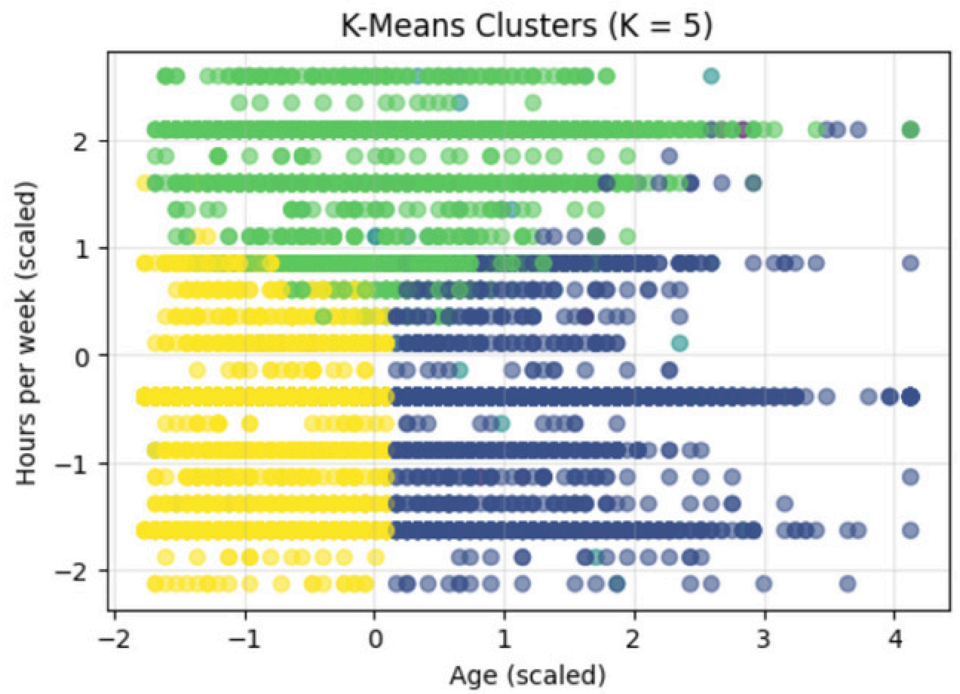
K = 3

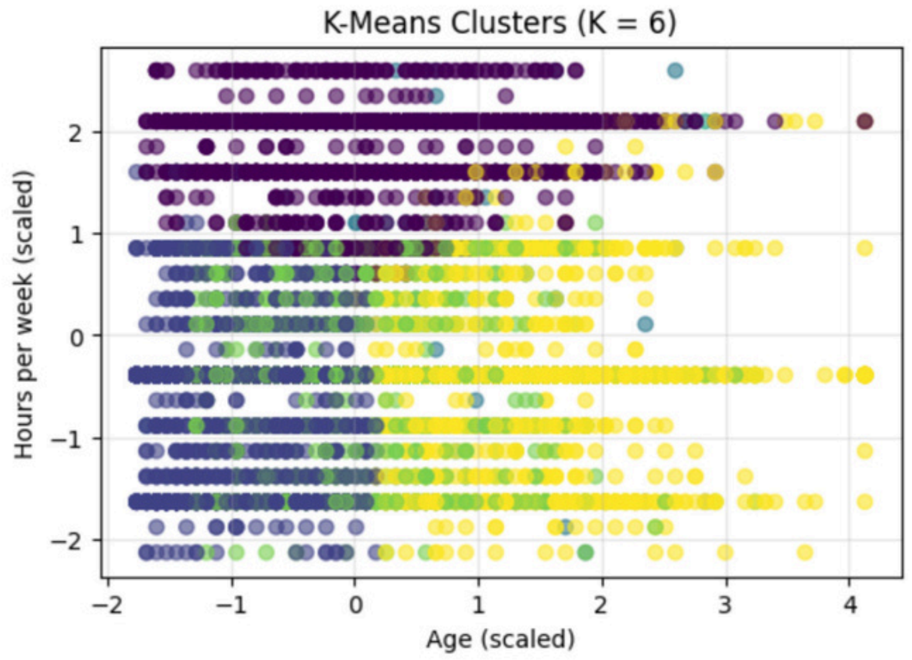


K = 4



K = 5





K = 6

CLUSTERING FINDINGS

K	Inertia (WCSS)	Silhouette Score
2	62706.43	**0.814**
3	51709.18	0.245
4	43066.76	0.252
5	29420.68	0.299
6	25530.55	0.307

The optimal number of clusters is **K = 3**, based on the majority rule (agreement between metrics).

clusters:

- 0 Younger adults, lower work hours, minimal capital gains.
- 1 Middle-aged adults, average education, moderate work activity
- 2 Older individuals, higher capital gains, longer work hours

SUMMARY OF FINDINGS

- Higher education + longer working = higher income.
- Younger adults = lower working hours + minimal capital gains.
- Experienced workers = more income potential.

SUMMARY

- Problem
- data
- Data preprocessing
- Data mining techniques
- Findings

Thank You